

Mohammad Ali Seyed Mahmoud

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Available September 2026 | 6–9 Month Full-Time Co-op

PROFILE

Mechanical Engineering Ph.D. researcher specializing in **computational materials engineering, finite-element modeling, probabilistic inference, and automated simulation workflows**. Experienced in developing validated ABAQUS/Python models that connect material parameters to experimentally measured mechanical response, including load–displacement and full-field measurements. Background includes FEM, Monte Carlo/Bayesian methods, materials characterization, microscopy, surrogate modeling, and high-throughput HPC simulation. Seeking to apply these transferable modeling and metrology capabilities to **semiconductor process and TCAD modeling**.

TECHNICAL SKILLS

- **Modeling & Simulation:** ABAQUS/Standard & Explicit; finite-element modeling (FEM); nonlinear mechanics; contact; large deformation; elastoplasticity; constitutive modeling; damage/fracture; parametric studies; mesh convergence and model verification.
- **Statistical & Computational Methods:** Bayesian inverse problems; Transitional Markov Chain Monte Carlo (T-MCMC); Gaussian-process (GP) surrogate modeling; PCA; simulation-based inference; Conditional Flow Matching; uncertainty quantification; active-learning methods.
- **Materials Characterization & Metrology:** Digital Image Correlation (DIC); SEM; optical microscopy; tensile, small-punch, and indentation testing; full-field displacement/strain analysis; mechanical property characterization; simulation–experiment correlation.
- **Programming & Automation:** Python (NumPy, SciPy, PyTorch); MATLAB; ABAQUS Python scripting; automated input generation and ODB post-processing; Slurm/HPC batch workflows; Git; data validation and scientific workflow automation.
- **Materials:** Aluminum alloys, steels, refractory alloys; microstructure–property–performance relationships; constitutive-property identification and model calibration.

COMPUTATIONAL MATERIALS & MODELING EXPERIENCE

Graduate Research Associate | Georgia Institute of Technology

Jan 2023 – Present

- **Finite-Element Materials Modeling:** Developed and validated ABAQUS models for metallic material characterization involving nonlinear contact, large plastic deformation, and damage, linking material parameters to experimentally measurable force and full-field deformation responses.
- **Bayesian / Monte Carlo Parameter Identification:** Developed a multimodal Bayesian calibration framework using force–displacement and DIC measurements, Gaussian-process surrogates, PCA, and T-MCMC to identify parameters of the Gurson–Tvergaard–Needleman damage model; work published in *Acta Materialia* (2026).
- **High-Throughput Simulation Automation:** Built Python-based pipelines for parametric ABAQUS studies, including automated input generation, job configuration, post-processing, feature extraction, and model validation across hundreds of finite-element simulations.
- **HPC Simulation Campaigns:** Developed Python/Slurm workflows for high-throughput ABAQUS execution on HPC systems and completed a 63-case expanded-domain simulation campaign with automated input-integrity checks, execution monitoring, ODB extraction, force/full-field validation, and reproducible provenance tracking.
- **Surrogate & Generative Modeling:** Developed a likelihood-free GP–Conditional Flow Matching framework for rapid estimation of elastic modulus and yield strength from multimodal small-punch-test measurements, enabling amortized posterior inference without repeated Monte Carlo sampling for each new specimen.
- **Experimental Metrology & Model Validation:** Integrated mechanical testing with DIC full-field measurements, microscopy, and computational models to quantify simulation–experiment agreement and identify material properties from limited-volume specimens.

Ongoing Ph.D. Research: Active Simulation-Based Inference

2026 – Present

- Developing a posterior-guided active-learning framework that adaptively selects expensive FE simulations to improve constitutive-property inference while maintaining global model accuracy and computational efficiency.
- Building a bounded autonomous computational workflow for candidate generation, ABAQUS execution, scientific validation, failure handling, surrogate retraining, uncertainty assessment, and reproducible decision/provenance tracking.

Graduate Research Assistant | University of Tehran

Sep 2020 – Dec 2022

- Developed Python/MATLAB computational methods for designing refractory-alloy microstructures with targeted thermal conductivity, connecting composition/microstructure descriptors with material performance; published in *Materials* (2023).

EDUCATION

Ph.D., Mechanical Engineering | Georgia Institute of Technology

Summer 2027

Novelis Graduate Scholar 2026 | GPA: 3.75/4.0

Relevant Coursework: Continuum Mechanics, Finite Element Methods, Elasticity, Plasticity, Computational Mechanics

M.Sc., Mechanical Engineering | University of Tehran

Jul 2022

Ranked Top 5% | GPA: 3.77/4.0

SELECTED PUBLICATIONS

- **M. A. Seyed Mahmoud**, D. Renner, A. Khosravani, and S. R. Kalidindi, “Sequential Bayesian Inference of the GTN Damage Model Using Multimodal Experimental Data.” *Acta Materialia*, 306, 121902, 2026.
- **M. A. Seyed Mahmoud**, A. Venkatraman, R. Mahat, S. Mitra, and S. R. Kalidindi, “Amortized Posteriors for Estimation of Material Constitutive Parameters from Multimodal Measurements on Small Punch Tests.” arXiv:2607.24534, 2026. *Under review*.
- **M. A. Seyed Mahmoud** et al., “Design of Refractory Alloys for Desired Thermal Conductivity . . .” *Materials*, 2023.